10.01.2023

Roll No. ....

TCE–301

B. TECH. (CIVIL ENGINEERING) (THIRD SEMESTER)
END SEMESTER EXAMINATION, Jan., 2023

MECHANICS OF FLUIDS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are twenty.

(iv) Each sub-question carries 10 marks.

1. (a) State Rayleigh method of dimensional analysis and explain procedure for the analysis. (CO1)

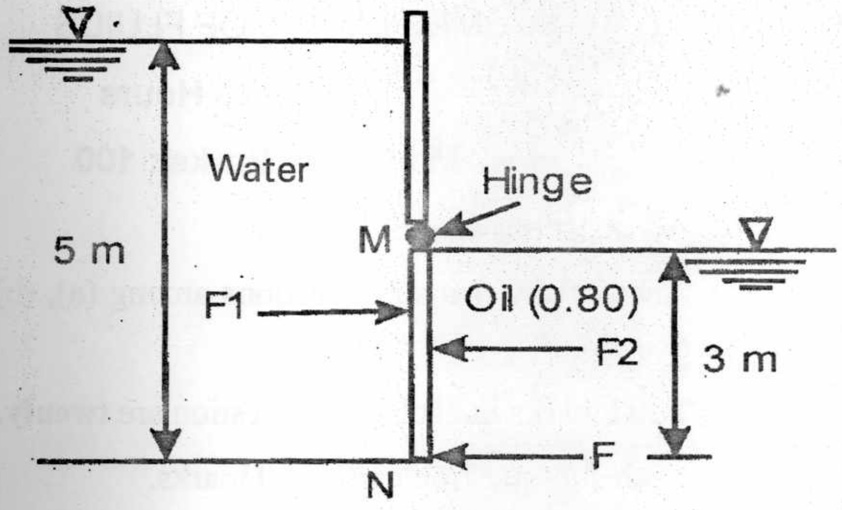
(b) A pump develops a power P which is a function of the discharge (Q), the head (H) and the specific weight (γ) of the fluid. Show that $P = KQ\gamma H$. Where, K is a dimensionless constant. (CO1)

(c) A plate of area 0.2 m^2 and 4 mm thickness is lifted through the midway between two large plates spaced 2.40 cm apart at a uniform speed of 12 cm/s. The gap is filled with oil of viscosity 1.80 Ns/m^2. The density of oil is 960 kg/m^3. The plate weighs 30 N. Determine the lifting force required. (CO1)

P. T. O.

2. (a) What do you mean by stability of submerged and floating bodies ? What are the various conditions of equilibrium of submerged and floating bodies ? (CO2)

(b) Figure shows a gate MN hinged at the end M. If the gate is 1.0 m wide, calculate the horizontal force required at N to keep the gate in equilibrium. (CO2)



(c) If the velocity component for two dimensional flows is given by $u = \dfrac{x}{x^2 + y^2}$ and $v = \dfrac{y}{x^2 + y^2}$, determine the acceleration components a_x, and a_y. (CO2)

3. (a) Obtain the expression for time of emptying a tank with no inflow provided with an orifice at the side of the tank. (CO3)

(b) What is Euler's equation of motion ? How will you obtain Bernoulli's equation from it ? (CO3)

(c) A rectangular channel 1.5 m wide has a discharge of 0.2 m^3/s, which is measured by a right angled V-notch. Find the position of the apex of the notch from the bed of the channel, if the maximum depth of water is not to exceed 1.0 m. Assume Cd= 0.62. (CO3)

4. (a) What is Stoke's law ? Derive an expression for terminal velocity. (C4)

(b) An oil of viscosity 4.25 poise flows in a 45 mm diameter pipe with a discharge of 5.25 lit./sec. If the specific gravity of the oil is 0.87, state whether the flow is Laminar or Turbulent. (CO4)

(c) Oil of mass density 800 kg/m^3 and dynamic viscosity 0.02 poise flows through 50 mm diameter pipe of length 500 m at the rate of 0.19 liter/s. Determine (i) Reynolds number of flow, (ii) Maximum velocity, (iii) Pressure gradient, (iv) Loss of pressure in 500 m length and (v) wall shear stress. (CO4)

5. (a) What do you mean by Pipe network ? Explain the Hardy Cross method of solving the pipe network problems. (CO5)

(b) What is meant by water hammer ? Derive water hammer equation due to gradual closure of valve. (CO5)

(c) A piping system consists of 3 pipes arranged in series; the lengths of the pipes are 1200 m, 750 m and 600 m and diameters 750 mm, 600 mm and 450 mm respectively. Transform the system to an equivalent 450 mm diameter pipe. Also, determine an equivalent diameter for the pipe, 2250 m long. (CO5)

Roll No. ....

TCE–302

B. TECH. (CE) (THIRD SEMESTER)

END SEMESTER EXAMINATION, Jan., 2023

BASIC SURVEYING

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are twenty.

(iv) Each sub-question carries 10 marks.

1. (a) Explain the various types of accessories used in chaining ? Mention the various types of tapes used in surveying. (CO1)

(b) A 30 m steel tape was standardized on flat ground at a temperature of 20°C and under a pull of 5 kg. The tape was used in the field at a temperature of 25°C and under a pull of 5 kg. The cross sectional area of tape is 0.02 cm^2, and its total weight is 660 g. The Young's modulus and coefficient of linear expansion of steel are 2.0×10^{6} kg/cm^2 and 11×10^{6} per °C respectively. Find the correct horizontal distance. (CO1)

P. T. O.

(c) Define surveying. Discuss in brief the principles of surveying. Explain the various classifications of survey. (CO1)

2. (a) The following are the bearings of a closed traverse : (CO2)

Side	FB	AB
AB	N 45°30′ E	S 45°30′ W
BC	N 60°0′ E	N 60°0′ W
CD	S 10°30′ W	N 10°30′ E
DA	N 75°45′ W	S 75°45′ E

Calculate the interior angles of the traverse.

(b) Explain the difference between prismatic compass and surveyor compass. (CO2)

(c) Explain instrumental, personal and natural error in theodolite surveying. (CO2)

3. (a) Calculate the latitudes, departures and closing error for the following traverse and adjust using bowditch's rule : (CO3)

Line	Length, m	WCB
AB	89.31	45°10′
BC	219.76	72°05′
CD	151.18	161°52′
DE	159.10	228°43′
EA	232.26	300°42′

(b) Explain the principle of stadia method and derive tacheometric distance equation. (CO3)

(c) Determine the horizontal distance between the instrument station P and the staff station Q from the following data : (CO3)

Height of the instrument= 1.490 m

Vertical angle = +4°30′

Staff reading (Staff vertical) = 0.645, 0.998, 1.351 m

Also, determine the R.L. of Q if that of P is 200.410 m. Take K= 100 and C= 0.0.

4. (a) Explain the different types of levelling operations. (CO4)

(b) The following consecutive readings were taken with a auto level along a chain line at a common interval of 15 m. The first reading was at a chainage of 165 m where the reduced level is 98.085. The instrument was shifted after the fourth and ninth readings. 3.150, 2.245, 1.125, 0.860, 3.125, 2.760, 1.835, 1.470, 1.965, 1.225, 2.390 and 3.035 m. Mark rules on page of your answer sheet in the form of a level book page and enter on it the above readings and find the reduced level of all points by the collimation method. (CO4)

(c) Define the terms 'contour line', 'contour interval'. State the uses of contour maps. (CO4)

5. (a) Discuss the errors that may occur in plane tabling. (CO5)

(b) Define orientation. Describe the methods of orientation with a sketch. (CO5)

(c) Discuss the advantage and disadvantage of plane table surveying over other methods. (CO5)

Roll No. ....

TCE–303

B. TECH. (CE) (THIRD SEMESTER)

END SEMESTER EXAMINATION, Jan., 2023

BUILDING MATERIALS AND CONSTRUCTION TECHNOLOGY

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Write down the characteristics of good bricks. Mention the size of nominal and standard brick. Explain the procedure of "manufacturing of brick" in the form of flow chart. (CO1)

   (b) Write down different methods of seasoning of timber along with their advantages and disadvantages. (CO1)

   (c) Write short notes on the following : (CO1)

   (i) Defects in Timbers

   (ii) Types of Mortar

2. (a) Mention the BOGUE compounds of cement and their role in providing strength to cement during the hydration process. (CO2)

P. T. O.

(b) Write short notes on the following : (CO2)

(i) High Alumina Cement

(ii) Portland Slag Cement

(c) Briefly discuss the following laboratory tests on cement : (CO2)

Consistency, Initial and Final setting time.

3. (a) Mention the names of different types of buildings as per NBC. Provide definition of Residential Building Education Building. (CO3)

(b) Write a detailed note on the various factors governing the planning of building. (CO3)

(c) Provide detailed discussion on the ventilation of building and why it is important to provide good ventilation in a building. (CO3)

4. (a) Provide comparative statements of Flat Roof and Sloping Roof in terms of their merits and demerits. (CO4)

(b) Discuss Damp Proof Course (DPC) in terms of cause and effect of dampness and suggested remedies. (CO4)

(c) Describe different types of staircase with diagram. (CO4)

5. (a) Differentiate Lift with Excalator with diagram. Explain why you feel heavy when the lift goes up and feel lighter when the lift comes down. (CO5)

(b) Briefly discuss paint in terms of characteristics of idea paint, constituents of paint and various defects in painting. (CO5)

(c) Write a detailed note on types of scaffolding with diagram. (CO5)

Roll No. ....

TCE–304

B. TECH. (CE) (THIRD SEMESTER)
END SEMESTER EXAMINATION, Jan., 2023

STRENGTH OF MATERIALS

Time : Three Hours

Maximum Marks : 100

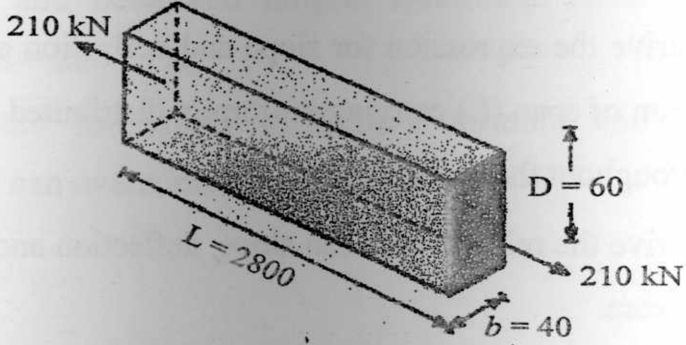
Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are twenty.

(iv) Each sub-question carries 10 marks.

1. (a) A 2.8 m long member is 60 mm deep and 40 mm wide in cross section. It is subjected to axial tensile force of 210 kN as shown in figure. Determine change in length, change in cross sectional dimensions. Assume E= 200 GPa and $\mu = 0.3$. (CO1)



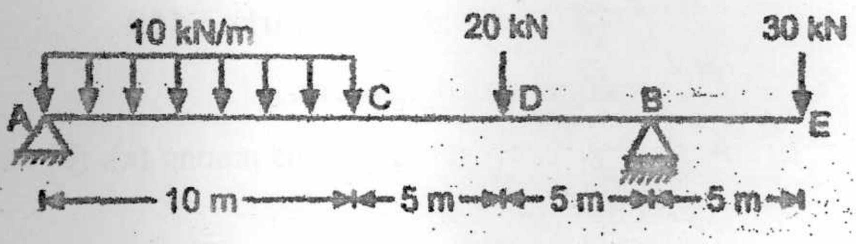
(All dimensions in mm)

P. T. O.

(b) Derive the expression for elongation of a bar of length (l) and uniform diameter (d) due to self-weight which is hanging vertically downward at one end. (CO1)

(c) A vertical bar of 20 mm diameter, 2 m long is fixed at top and supports a downward load of 20 kN at bottom. Calculate elongation of bar assuming density = 78 kN/m^3 and E=200 GPa. (CO1)

2. (a) Draw SFD and BMD for given beam indicating all important values. (CO2)



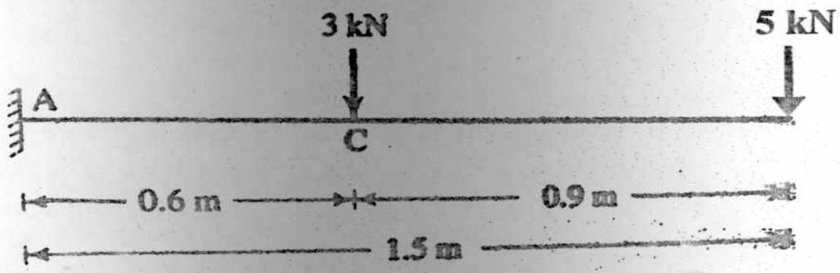
(b) Explain the theory of simple bending with assumption and derive the bending equation. (CO2)

(c) A wooden beam supports UDL of 40 kN/m run over a simply supported span of 4 m. It is of rectangular cross-section, 200 mm wide and 400 mm deep. Sketch the shear stress distribution, and determine (i) maximum shear stress (ii) average shear stress. (CO2)

3. (a) Derive the expression for slope and deflection at free end of cantilever beam of span (L) carrying uniformly distributed load (w per unit length) throughout the span. (CO3)

(b) Derive the relation between slope, deflection and radius of curvature for a beam. (CO3)

(c) Determine the slope and deflection at free end of a cantilever shown in figure. The section is 0.3 m × 0.4 m. E = 12 GPa. (CO3)



4. (a) Prove that shear stress along any radius varies linearly with radial distance of the shaft. (CO4)

(b) Find the power transmitted by a shaft having 60 mm diameter rotating at 150 rpm, if maximum permissible stress is 80 MPa. (CO4)

(c) Define the terms : torsional stiffness, torsional flexibility, torsional rigidity and torsional section modulus. (CO4)

5. (a) A cylindrical shell is 3 m long, having 1 m internal diameter and 15 mm thickness. Calculate maximum intensity of shear stress induced and also change in diameter of shell if it is subjected to internal fluid pressure of 1.5 N/mm^2. (CO5)

(b) A square column of 100 mm × 100 mm cross section has a concentric longitudinal hole of 50 mm diameter. The length of column is 5 m, one end fixed and other end hinged. Determine Euler's buckling load assuming E = 200 GPa. (CO5)

(c) Describe the limitations of Euler's formula and how Rankine's hypothesis can overcome this limitation. (CO5)

80

Roll No. ....

TCE–305

B. TECH. (THIRD SEMESTER)

END SEMESTER EXAMINATION, Jan., 2023

ENGINEERING MECHANICS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

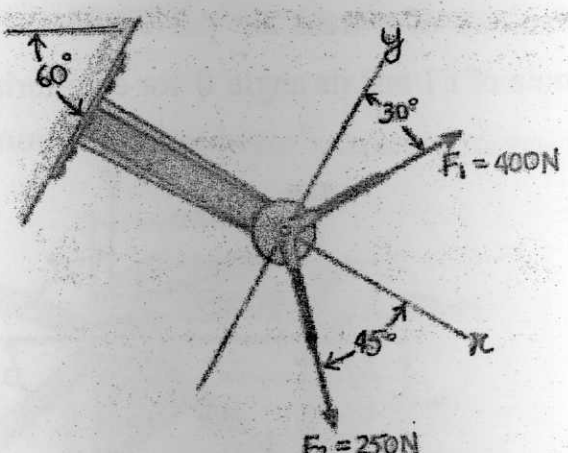
(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are twenty.

(iv) Each sub-question carries 10 marks.

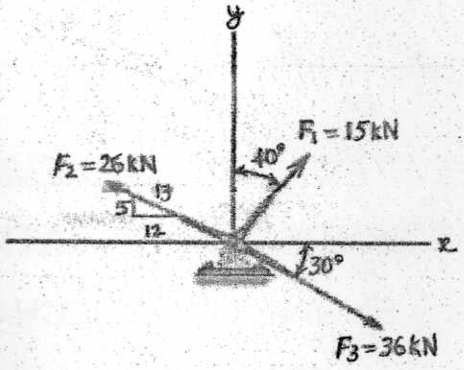
1. (a) State Newton's three laws of motion with examples. (CO1)

(b) Resolve each of the forces into its x and y components. (CO1)



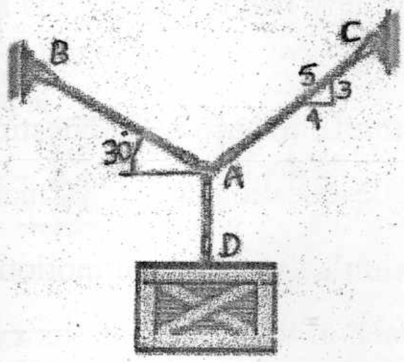
P. T. O.

(c) Determine the magnitude of the resultant force and its direction, measured counter-clockwise from the positive x-axis. (CO1)

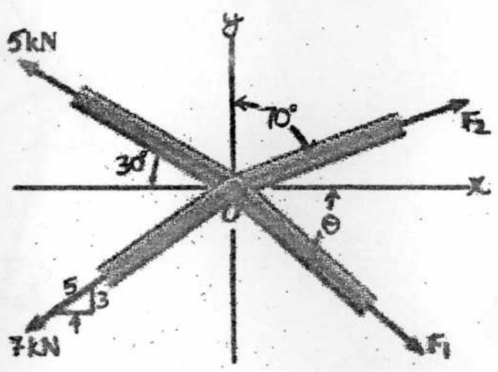


2. (a) Define a force and explain its classification. (CO2)

(b) The crate has a weight of 550 lb. Determine the force in each supporting cable. (CO2)

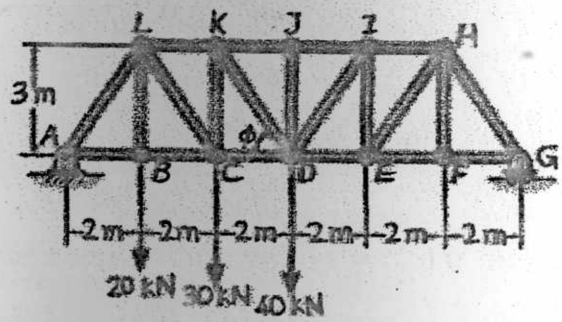


(c) The members of a truss are pin connected at joint O. Determine the magnitude of F1 and its angle θ for equilibrium. Set F2 = 6 kN. (CO2)

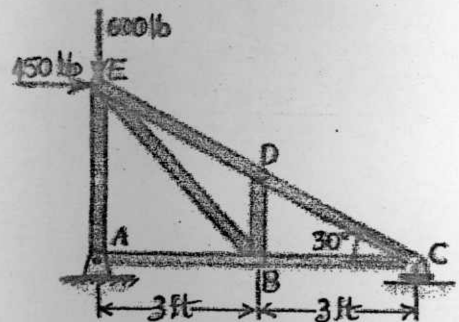


3. (a) What is a beam ? Explain different types of beams and supports with diagrams. (CO3)

(b) Determine the force in members KJ, KD and CD of the Pratt truss. State if the members are in tension or compression. Use method of sections. (CO3)

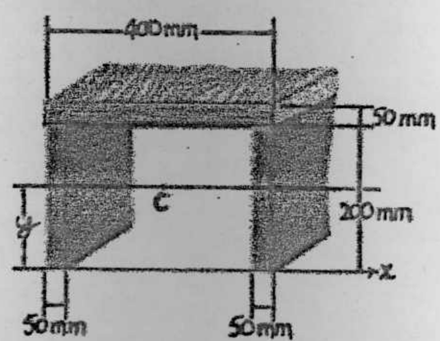


(c) Determine the force in each member of the truss. State if the members are in tension or compression. (CO3)

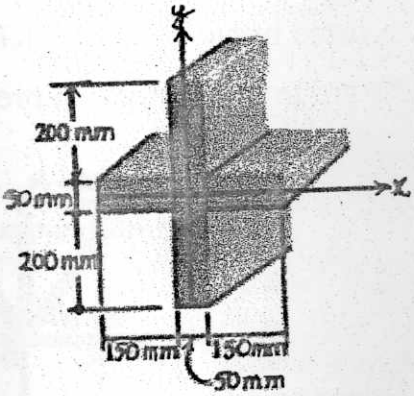


4. (a) Explain centroid, moment of inertia of an area and radius of gyration of plane figures. (CO4)

(b) Locate the centroid y of the beam's cross-sectional area. (CO4)

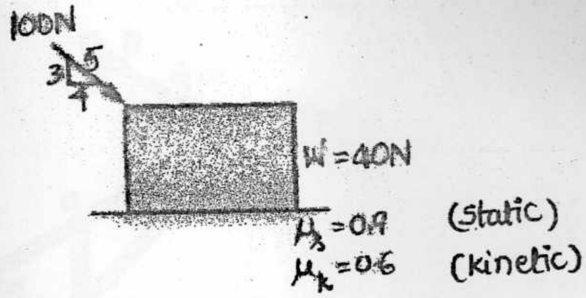


(c) Determine the moment of inertia of the beam's cross-sectional area about the centroidal x and y axes. (CO4)

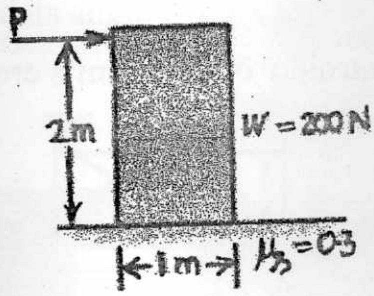


5. (a) Explain the Laws of static friction. Define angle of repose and angle of friction. (CO5)

(b) Determine the friction force at the surface of contact. (CO5)



(c) Determine the force P needed to cause impending motion of the block. (CO5)



Roll No. ....

TMA–302

B. TECH. (CIVIL ENGG.) (THIRD SEMESTER)

END SEMESTER EXAMINATION, Jan., 2023

ENGINEERING MATHEMATICS—III

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are twenty.

(iv) Each sub-question carries 10 marks.

1. (a) Find the Fourier cosine transform of the function : (CO2)

$$f(x) = \begin{cases} x^2, & 0 \le x \le a \\ 0, & x \ge a \end{cases}$$

(b) Find the Fourier sine integral for the function $f(x) = e^{-kx}; x \ge 0$, where k is any positive constant. Hence show that $\int_0^\infty \frac{\sin \lambda x}{k^2 + \lambda^2} d\lambda = \frac{\pi}{2} e^{-kx}$. (CO2)

(c) Express the function $f(x)$ as a Fourier integral representation and hence evaluate : (CO2)

$$\int_0^\infty \frac{\sin \lambda \cos \lambda x}{\lambda} d\lambda,$$

where $f(x) = \begin{cases} 1, & \text{if} \mid x \mid \le 1 \\ 0, & \text{if} \mid x \mid > 1 \end{cases}$.

P. T. O.

2. (a) Show that the function $f(z) = u + iv$, where : (CO1)

$$f(z) = \begin{cases} \dfrac{x^2y^5(x+iy)}{x^4+y^{10}}, & z \neq 0 \\ 0 \quad , & z = 0 \end{cases}$$

is not analytic at $z = 0$, although Cauchy-Riemann equations are satisfied at $z = 0$.

(b) Find the values of α and β such that the function : (CO1)

$$f(z) = (x^2 + \alpha y^2 - 2xy) + i(\beta x^2 - y^2 + 2xy)$$

is analytic. Also find $f'(z)$ in terms of x and y.

(c) Using Cauchy's integral formula to evaluate the value of the integral $\oint \frac{e^{zt}}{z^2+1} dz$, along the closed curve $C : |z| = 3$. (CO1)

3. (a) Using Newton's method, find a real root of an equation $3x = 1 + \cos x$ correct to three decimal places. (CO3)

(b) The velocity (v) of a particle at distance (s) from a point on its path is given by the following table : (CO3)

Distance (s) (metres)	**Velocity (v) (m/sec)**
0	47
10	58
20	64
30	65
40	61
50	52
60	38

Calculate the time taken to travel 60 metres by using Simpson's one-third rule.

(c) Find a real root of the equation $x = e^{-x}$ by using Bisection method correct to three decimal places. (CO3)

4. (a) A random variable X has the following probability distribution : (CO6)

$X = x$	$P(X = x)$
-2	0.1
-1	k
0	0.2
1	$3k$
2	$2k$
3	0.3

Determine k and compute $P(X < 2)$.

(b) A sample of 100 battery cells is tested to find length of life with mean 10 hours and standard deviation 4 hours. Assuming that the data is normally distributed, what percentage of battery cells are expected to have life : (i) more than 13 hours, (ii) less than 5 hours ? (CO6)

(c) If the probability density function of a continuous random variable X is given by :

$$f(x) = \begin{cases} kx^2; & 0 < x < 1 \\ 0; & \text{otherwise} \end{cases}$$

Find the value of k and hence compute its mean, variance and standard deviation. (CO6)

5. (a) The following are the values of two variables, diastolic blood pressure (y) and time in minutes (x). Diastolic blood pressure (y) was recorded at various time periods (x) on individuals. Find the regression line of y on x and x on y. Also find the coefficient of correlation : (CO4 & 5)

Time periods (x)	Diastolic blood pressure (y)
0	72
5	67
10	70
15	65
20	66

(b) Following data depicts the statistical values of rainfall and production of wheat in a region for a specified time period : (CO4 & 5)

	Mean	Standard Deviation
Production of Wheat (kg/unit area)	10	8
Rainfall (cm)	8	2

Estimate the production of wheat when rainfall is 9 cm if correlation coefficient between production and rainfall is given to be 0.5.

(c) The first moments about the working mean 28.5 of a distribution are 0.294, 7.144, 42.409 and 454.98. Calculate the moments about the mean. Also calculate β_1, β_2 and comment upon the Skewness and Kurtosis of the distribution. (CO4 & 5)

80